DATA 412/612 · Statistical Programming in R · Week 3

# 3.1 ggplot2: The Grammar of Graphics

A plot is data, a mapping, and a geom. ggplot() is the canvas; + adds a layer.

## 1. Why we plot

Identical summary statistics do not imply identical distributions. Anscombe's quartet consists of four data sets with the same means, variances, and correlation. The scatterplots differ: one is linear, one is curved, one is linear with an outlier, and one is a vertical stack with an outlier. The Datasaurus dozen demonstrates the same point. Thirteen data sets share one set of summary statistics and display different shapes, including a Tyrannosaurus rex.

A plot is used both during exploration and when a result is communicated. Decoration that does not encode a variable should be omitted.

R has three plotting systems:

- Base R: start a blank plot, then add pieces (`plot()`, `lines()`, `rug()`)
- `lattice`: one large function call
- `ggplot2`: layers that follow a grammar of graphics

This course uses `ggplot2`. It comes with `library(tidyverse)`. You do not load it again.

```
library(tidyverse)
data("mpg")
```

`mpg` is a tibble of cars: engine size (`displ`), city and highway mpg (`cty`, `hwy`), drive (`drv`), class, and more. Type `?mpg`.

A Base R scatterplot with a loess curve and rugs takes several unconnected calls. The ggplot2 version is one chain:

```
ggplot(mpg, mapping = aes(x = displ, y = hwy)) +
  geom_point() +
  geom_smooth(se = FALSE, method = loess) +
  geom_rug() +
  xlab("Displacement") +
  ylab("Highway")
```

`qplot()` is a short ggplot2 helper for a quick look. We will write `ggplot()` so the layers stay visible.

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## 2. The grammar, in four pieces

The grammar of graphics treats a plot as layers. You can get most of the work done with four of them:

- **data**: a data frame
- **aesthetic mappings**: which column is `x`, `y`, `color`, `shape`, `size`, `alpha`, `fill`, ... Created with `aes()`
- **geoms**: the marks, `geom_point()`, `geom_histogram()`, `geom_boxplot()`, ...
- **facets**: a panel per level of a categorical variable, `facet_wrap()` or `facet_grid()`

There are also stats (binning for a histogram), positions (jitter, dodge, stack), and coordinate systems. We will change those when we need them.

`ggplot()` makes the canvas. It needs `data` (a data frame) and, usually, `mapping = aes(...)`.

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy))
```

That draws axes and an empty panel. There is no geom yet.

Add a layer with `+` at the **end** of the line, then the next function on the next line. A `+` at the start of a line will error.

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy)) +
  geom_point()
```

You can pipe the data in. The first argument of `ggplot()` is then filled, so you only write the mapping:

```
mpg |>
  ggplot(mapping = aes(x = displ, y = cty)) +
  geom_point()
```

Larger `displ` goes with smaller `cty`: a negative relationship.

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## 3. Which variable is `x`

You choose the axes from the question. The response usually goes on `y`. The explanatory variable goes on `x`.

"Plot A vs B" and "plot A as a function of B" both put B on `x` and A on `y`. Swapping the axes does not change the association. It does change what a reader thinks is doing the explaining.

```
ggplot(data = mpg, mapping = aes(x = cty, y = hwy)) +
  geom_point()
```

`cty` and `hwy` move together: a positive relationship.

## 4. A third variable: categorical

A categorical variable has a short list of groups. In `mpg`, `drv` is `f` (front), `r` (rear), or `4`.

Map it to `color` (usually clearer) or `shape` (harder once you have many levels):

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy, color = drv)) +
  geom_point()

ggplot(data = mpg, mapping = aes(x = displ, y = hwy, shape = drv)) +
  geom_point()
```

Fifty states as fifty colors is a mess. Recode into fewer groups, or facet (note 3.3).

`size` and `alpha` have a natural order (small to large, faint to dark). A categorical variable usually does not. Mapping `drv` to `size` or `alpha` is legal and usually a bad idea.

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## 5. A third variable: quantitative

A third number can go on `color` (a spectrum), `alpha` (transparency), or `size` (a bubble chart; worst when points overlap).

```
ggplot(data = mpg, mapping = aes(x = displ, y = hwy, color = cty)) +
  geom_point()

ggplot(data = mpg, mapping = aes(x = displ, y = hwy, color = cty)) +
  geom_point() +
  viridis::scale_color_viridis()

ggplot(data = mpg, mapping = aes(x = displ, y = hwy, alpha = cty)) +
  geom_point()

ggplot(data = mpg, mapping = aes(x = displ, y = hwy, size = cty)) +
  geom_point()
```

`shape` has a short list of symbols. Mapping a continuous variable to `shape` errors.

Stacking `color`, `size`, `alpha`, and `shape` at once is possible and usually unreadable. If you cannot say the sentence the plot is making, drop a mapping.

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## 6. Practice

1. Load `iris` (`data(iris)`; `head(iris)`). Which columns are quantitative? Which is categorical?
2. Scatterplot petal length vs petal width, colored by species. What do you see?

3. Map a categorical column in `mpg` to `size` or `alpha`. Why is that a poor default?